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ARcenso: Expanding and Strengthening an R Package for Argentina's Census Data

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Executive Summary

ARcenso is an open-source R package that provides reproducible access to Argentina's official census statistics. Developed through the rOpenSci Champions Program, the package already provides historical census data in a documented and reusable format, but expanding its coverage and strengthening its technical infrastructure requires substantial development work. Historical census results are scattered across printed publications, scanned documents, and spreadsheets whose structures and classifications vary across census rounds, making them difficult to access, compare, and reuse. This proposal seeks funding to support the next stage of ARcenso's development and strengthen it as sustainable technical infrastructure for working with these data.

The project will focus on strengthening ARcenso and preparing it for its first release on CRAN and publication on R-universe. This will include improvements to the package's internal infrastructure, including data storage and data types, existing functionality, testing, and documentation, so that it can support the incorporation of additional census data. Building on this infrastructure, census coverage will be progressively expanded with the 1960 census and national and subnational data from the 1980 and 1991 censuses, using a reproducible and documented processing and validation workflow.

The proposed work will be carried out over nine months, from January to September 2027. We request USD 5,500 to support 245 hours of team work. The resulting package will provide statistical offices and public agencies, international and regional organizations, researchers and professionals, universities, and other users of population data with a more robust and reproducible resource for accessing and analyzing Argentina's historical census statistics. At the same time, the project will

contribute to the R ecosystem and provide a practical example of how open-source R infrastructure can support the preservation and reproducible use of historical official statistics in Latin America and other countries facing similar challenges.

Signatories

Project team

Andrea Gomez Vargas - Sociologist and Population Statistics Analyst

Andrea holds a degree in Sociology and is currently pursuing a Master's degree in Generation and Analysis of Statistical Information. She specializes in population and official statistics and works at Argentina's National Institute of Statistics and Censuses (INDEC), where she has experience in the production and analysis of population statistics, including work on the final results of Argentina's 2022 Population, Household and Housing Census. She is the main author of ARcenso, an open-source R package initiated through the rOpenSci Champions Program. Her contributions to the project combine expertise in census data and statistical methodology with R package development, particularly in the identification, harmonization, validation, and documentation of historical census data.

Emanuel Ciardullo - Statistician and Data Science Coordinator

Emanuel holds a degree in Statistics and is the coordinator of the Data Science and Alternative Data Sources Integration area at Argentina's National Institute of Statistics and Censuses (INDEC). His experience includes statistical methodology, household survey weighting, time-series analysis, and seasonal adjustment. He has also provided international technical assistance in statistics through World Bank projects. He is a co-developer of CEMRepBoot, a statistical estimation and analysis tool developed in R and Shiny, and a co-author of ARcenso, where he contributes statistical, methodological, and technical expertise to data processing and package development.

Selected public artifacts

- [ARcenso repository](#)
- [ARcenso documentation](#)
- [rOpenSci Champions blog: From Census Data to Demographic Analysis with ARcenso: A Reproducible Workflow in R](#)
- [useR! 2025 - ARcenso: a Package Born From Chaos, Powered by Community](#)
- [LatinR 2024 - ARcenso: First Steps in Developing an R Package in Community with rOpenSci](#)
- [Other ARcenso outreach activities.](#)

The Problem

Argentina’s national population censuses are a key input for the production of official population and social statistics, and a fundamental source for demographic and social research. With national and subnational coverage spanning several decades, census data enable the analysis of long-term population and social change and provide evidence for research and public policy. However, historical census results are dispersed across printed publications, scanned documents, and spreadsheets, while their structure and classifications vary from one census round to the next. This makes the data difficult to access, compare, and reuse, and poses a challenge for preserving the country’s statistical memory.

The problem affects a broad community of users: statistical offices and other public agencies; international and regional organizations that rely on official population statistics; researchers and professionals in demography, the social sciences, economics, health, and other fields that use population data; university teachers and students working with official statistics; and journalists, civil society organizations, and other users interested in long-term demographic and social change. Working with historical data across census rounds can require locating original sources, transcribing or reshaping tables, interpreting changes in structures and classifications, and validating the results before they can be incorporated into reproducible workflows. These tasks are often repeated by individual users, creating duplicated effort and increasing the risk of inconsistencies or errors.

Existing resources provide access to census results in different ways, but they do not address their integration for reproducible use. Argentina’s National Institute of Statistics and Censuses (INDEC), the country’s national statistical office, makes official census results publicly available through its website, including individual tables generally distributed as separate Excel files and publications in PDF format. Starting with the census rounds of the 2000s, REDATAM was introduced in Argentina as an additional means of accessing census data. Developed by the United Nations Economic Commission for Latin America and the Caribbean (ECLAC), the software ([Economic Commission for Latin America and the Caribbean, n.d.](#)) allows users to generate customized tables through a web interface by selecting variables and levels of disaggregation.

In R, packages are available for working with REDATAM databases, but their use depends on access to those databases and follows a different model from ARcenso. IPUMS International ([IPUMS, n.d.](#)), in turn, provides harmonized census microdata samples for comparative research, but does not reproduce the official tables published by INDEC. Taken together, these resources do not provide the official published tables from Argentina’s different census rounds in a common, documented, and reusable structure.

ARcenso began addressing this gap through the rOpenSci Champions Program ([Ogundepo et al. 2024](#)) as an open-source R package that organizes Argentina’s official census tables and makes them directly available for reproducible analysis in R ([Gomez Vargas and Ciardullo 2026](#)). The package currently includes the 1970 census at the national and subnational level and the 1980 census at the national level, together with a public roadmap for progressively expanding its historical and geographic coverage. However, most of the census series has yet to be integrated, and its expansion requires substantial work in data recovery, harmonization, validation, and software development.

Solving this problem would provide a single, documented, and reproducible entry point to Argentina’s official census statistics directly from R. ARcenso would facilitate the consultation, processing, and longitudinal and comparative analysis of these data at the national and subnational levels, reducing the work required to integrate information across census rounds. In this way, it could serve as a resource for national and international public organizations, as well as for academic

research, teaching, and other fields that work with population data. At the same time, it would contribute to preserving historical statistical information in an open and reusable format.

The proposal

Overview

This proposal aims to expand and strengthen ARcenso as open-source R infrastructure for reproducible access to Argentina’s official census statistics. By the end of the project, ARcenso will have a stronger and more scalable technical infrastructure, supporting its first release on CRAN and distribution through R-universe, while expanding the census data available through the package with additional national and subnational tables from the 1960, 1980, and 1991 censuses.

The project is organized around three complementary lines of work:

- **Strengthening the package infrastructure and functionality:** We will continue strengthening ARcenso in line with rOpenSci package development guidelines and the technical requirements of CRAN and R-universe. Building on the package’s existing functions, we will improve and extend its functionality as census coverage grows, strengthen tests and documentation, and develop reproducible vignettes and workflows. We will also improve internal data storage, including appropriate data types and the consolidation and compression of data files where needed, and prepare the package for its first release on CRAN and R-universe.
- **Automating the data-processing workflow:** We will build on ARcenso’s existing R-based workflow for converting census source files into package data, incorporating automation strategies such as AI-assisted extraction and GitHub Actions. The workflow will remain reproducible and version-controlled, with automated validation and human review.
- **Expanding and harmonizing census coverage:** We will advance ARcenso’s public roadmap by completing the 1980 census at the subnational level and incorporating the 1991 census at the national and subnational levels. We will also add the 1960 census, whose digital tables were released by INDEC in August 2026. Together, this expansion represents approximately 3,500 Excel files containing official census tables that still need to be classified, standardized, documented, and validated.

ARcenso is already a functional open-source R package, used in teaching, university activities, and R community events. The project will run from January to September 2027. By strengthening the package infrastructure and its data-processing workflow, this project will provide the R community and other users of census data with a reproducible and sustainable resource for working with Argentina’s historical census statistics, while also supporting the teaching and learning of R through real-world data processing, visualization, and reproducible reporting. It will also establish the technical foundation for extending the roadmap to the 2001, 2010, and 2022 census rounds.

Detail

Minimum Viable Product

The minimum viable product is a strengthened release of ARcenso that:

- **implements the improved package infrastructure**, including a more scalable internal data storage structure and metadata for data provenance and validation;

- **strengthens the package functionality and quality**, maintaining the existing user-facing functions (`get_census()`, `check_repository()`, and `arcenso_app()`) while extending tests, documentation, and reproducible vignettes;
- **is prepared for its first release on CRAN and distribution through R-universe**; and
- **incorporates the 1991 national-level census tables** through the reproducible data-processing and validation workflow.

This release would deliver a more robust and maintainable package, extend the available census series to 1991, and establish the technical infrastructure needed to scale ARcenso to additional census rounds and subnational data.

Architecture

ARcenso connects official census sources with R users through two main technical layers: the internal data infrastructure of the package and the reproducible development pipeline used to process and validate the data. Users access the resulting census data through the package’s existing functions (`get_census()`, `check_repository()`, and `arcenso_app()`).

Data infrastructure. Currently, the package stores about 390 individual `.rds` files, with numeric values stored as text. To support expanded census coverage while keeping the package maintainable, the internal data structure will:

- store numeric values as integer or double columns;
- consolidate tables into a smaller number of compressed files, rather than storing one file per table;
- extend `census_metadata` with the source of each table (publication and table number) and its validation status; and
- keep these internal changes transparent to users through the existing package interface.

The architecture is designed to support the expanded census coverage while remaining suitable for distribution through CRAN and R-universe. If the complete data collection cannot remain within CRAN’s size limits, the data can be separated into a companion data package distributed through R-universe, while the main ARcenso package is submitted to CRAN ([Anderson and Eddelbuettel 2017](#)).

Development pipeline. ARcenso’s existing R-based workflow will be extended into a reproducible and version-controlled pipeline for processing and validating census sources. Development files and scripts will remain in `data-raw/` and will not be included in the built package. The workflow will follow these steps:

1. **Source and classification.** Official census source files are identified, classified, and registered with their references and metadata.
2. **Processing and standardization.** Versioned R scripts transform source tables into the standardized structure used by ARcenso. Where useful, AI-assisted extraction can support this process, but the resulting transformation and data remain reproducible without requiring AI tools.
3. **Automated validation.** Automated checks verify published totals and subtotals, data types, expected structures, and consistency across related tables.
4. **Human review.** Processed tables and validation results are reviewed before being incorporated into the package.

The pipeline will rely primarily on R, consistent with the package's current approach of minimizing dependencies.

Assumptions

No critical assumptions have been identified that would invalidate the project. The proposed infrastructure improvements are designed to deliver value independently of the final number of census tables incorporated during the project period.

External dependencies

- **R packages.** ARcenso depends on R ($\geq 4.1.0$) and imports `gt`, `shiny`, and `tidyselect`. Tests use `testthat`, and vignettes use `knitr` and `rmarkdown`. The complete and current list of package dependencies is maintained in the [DESCRIPTION](#) file.
- **Data sources.** Official census tables and publications produced by Argentina's National Institute of Statistics and Censuses ([INDEC](#)) for the census rounds included in the project.
- **Development and distribution infrastructure.** GitHub is used for version control and collaborative development, Codecov for test coverage, pkgdown for package documentation, Zenodo for archiving releases, and CRAN and R-universe for package distribution.

Failure modes and recovery

The main technical failure mode identified is that the expanded census collection may become too large to be efficiently stored and distributed within the main ARcenso package. If this occurs, the census data will be separated from the main package and made available through a companion data package or another R-accessible data distribution mechanism, while maintaining ARcenso as the user-facing interface.

Project plan

Start-up phase

ARcenso already has the basic infrastructure in place, allowing the team to begin work on technical deliverables from the first month:

- **Collaboration platform.** The code is hosted in a public [GitHub repository](#), with a bilingual (English and Spanish) contributing guide, a code of conduct, continuous integration with R CMD check, and test coverage reporting. The package has also undergone a technical review to identify areas for improvement in R package development good practices. During the start-up phase, we will add issue templates for reporting errors in census tables and requesting new tables.
- **License.** ARcenso is released under the MIT license, which will be maintained for all code produced during the project, including the data-processing pipeline.
- **Reporting framework.** The team will hold bi-weekly check-ins to review progress, while project activities and deliverables will be tracked through the public GitHub repository. A short progress report will be prepared every three months, shared with the ISC and made publicly available through the project repository, summarizing completed deliverables, progress in census coverage, and any relevant changes to the project plan.

Technical delivery

The project will run for nine months, from January to September 2027, in four phases:

Phase	Period	Deliverables
1. Package infrastructure	January – March 2027	Address package quality improvements identified through <code>pkgcheck</code> , while maintaining a clean R CMD <code>check</code> ; restructure internal data storage and data types; and review and adapt existing functions in preparation for CRAN and R-universe distribution.
2. Processing and validation pipeline	April 2027	Develop a reproducible data-processing and validation workflow; document the review process; and incorporate the 1991 national-level census tables as the first application of the workflow.
3. Census coverage	May – August 2027	Continue the incorporation, standardization, documentation, and validation of the 1960 census and subnational tables from the 1980 and 1991 censuses using the package infrastructure and processing pipeline.
4. Testing, documentation and distribution	September 2027	Finalize package testing and technical refinements; update documentation and vignettes; conduct community testing; and prepare ARcenso for its first CRAN release and publication on R-universe.

Key target dates:

- **31 March 2027:** Package infrastructure improvements completed, including internal data storage and data types, package quality improvements, and review of existing functions.
- **31 August 2027:** Data-processing and validation pipeline implemented and applied to the planned expansion of census coverage.
- **30 September 2027:** Final project delivery, including package testing, updated documentation and vignettes, community testing, the first CRAN release, and publication on R-universe.

Other aspects

All code produced through the project will remain openly developed on GitHub under the MIT license, with contributions welcome in both English and Spanish. ARcenso will be prepared for distribution through CRAN and R-universe, and package releases will continue to be archived on Zenodo with a DOI.

To make the project's progress and outputs publicly accessible and support the adoption and dissemination of ARcenso, we plan to:

- **Public project progress.** Maintain a dedicated public section in the ARcenso project documentation to track grant progress, deliverables, and related outputs.
- **R Consortium blog.** Share quarterly updates through the R Consortium blog, documenting ARcenso's technical development, project milestones, and outcomes.
- **Local engagement and communities.** Lead and organize in-person activities in Buenos Aires for R users and others interested in working with census data, together with R en Buenos Aires, RLadies+ Buenos Aires, universities, and other relevant communities and institutions.
- **Conferences.** Submit talks about ARcenso to LatinR and useR! , and other relevant conferences during 2027–2028, sharing the package and the technical experience of developing reproducible infrastructure for historical census data.
- **Social media and R community channels.** Share project progress, releases, and opportunities for use and contribution through social media and relevant R community channels.

Budget & funding plan

The requested budget is **USD 5,500** and covers labor only, for a total of **245 hours of team work**, at an effective rate of approximately **USD 22.45 per hour**. The budget is allocated across the four phases described in the Technical delivery section:

Phase	Period	Team hours	Estimated cost (USD)
1. Package infrastructure	January – March 2027	70	\$1,571
2. Processing and validation pipeline	April 2027	40	\$898
3. Census coverage	May – August 2027	95	\$2,133
4. Testing, documentation and distribution	September 2027	40	\$898
Total	January – September 2027	245	\$5,500

In the event of partial funding, priority will be given to the package infrastructure and the reproducible processing and validation pipeline, as these provide the technical foundation for the continued expansion of ARcenso beyond the grant period.

Success

Definition of done

The project will be considered successful when ARcenso provides a stable release that:

- **implements the planned package infrastructure improvements**, including improvements identified through `pkgcheck`, a more scalable internal data storage structure, and appropriate data types;
- **implements and documents a reproducible data-processing and validation pipeline**, demonstrated through the incorporation of the 1991 national-level census tables;
- **expands the package's census coverage** using the new infrastructure and processing workflow;
- **maintains and strengthens the existing package functionality**, including `get_census()`, `check_repository()`, and `arcenso_app()`, as census coverage expands;
- **includes updated tests, documentation, and reproducible vignettes** for the expanded package; and
- **is released on CRAN and published on R-universe.**

Measuring success

Progress and success will be measured through:

- **Intermediate package releases and NEWS.md**, documenting technical improvements, expansion of census coverage, and implementation of the processing and validation pipeline throughout the project;
- **quarterly R Consortium blog updates**, providing a public record of project progress, milestones, and outcomes; and
- **the final stable release**, with a clean R CMD check, updated documentation and vignettes, the first CRAN release and publication on R-universe.

Future work

After this project, the remaining stages of the ARcenso roadmap will include the 2001, 2010, and 2022 censuses at the national and subnational levels. The technical infrastructure and reproducible processing and validation pipeline developed through this project will provide the foundation for incorporating these census rounds in future releases.

Other possible extensions include:

- submitting ARcenso to rOpenSci software peer review;
- exploring tools for harmonizing selected variables across census rounds to facilitate longitudinal comparisons;
- expanding ARcenso to incorporate other census-related demographic data products, such as official population projections and mortality tables; and
- documenting the processing pipeline as a reusable workflow for other projects working with historical statistical publications.

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